PROPOSAL 2

JUST Student Assistant

Phase 2 — Project Proposal and Evaluation Report

GitHub: github.com/hibanaimat05/JUST-Student-Assistant

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1. Team and Roles

Team Member	Student ID	Role
Rand Al-Darabseh	174269	Product Lead
Batool Abualhayjaa	168361	Data Lead
Hiba Al-Naimat	170251	Evaluation Lead & Engineering Lead

2. User and Problem

JUST students use multiple university digital services for email, Wi-Fi, e-learning, mobile applications and IT support. Finding accurate instructions across different sources can be time-consuming, especially when students need clear step-by-step guidance. The project addresses this problem by developing a university-focused assistant that answers student questions using institutional support materials.

Project objective: build and evaluate an assistant that can provide relevant, clear and source-grounded answers to common JUST student support questions.

3. Data

The knowledge base was prepared from university support materials. PDF documents were converted to text, OCR was applied where needed, and the extracted content was cleaned and deduplicated. The documented document split contains 38 documents: 34 training documents (90%) and 4 held-out documents (10%), using random_state=42. The split manifest is stored at data/split_90_10/split_manifest.csv.

The final supervised fine-tuning dataset contains 304 valid question-answer records, split into 273 training records and 31 validation records. A separate evaluation set contains 30 test questions with reference answers. The document split and Q/A split are distinct: 34/4 refers to source documents, while 273/31 refers to SFT records.

4. Training and System Design

Base model: Qwen2.5-0.5B-Instruct, selected as the foundation model with instructor approval recorded in the project history.

Continued Pre-Training (CPT): LoRA-based CPT was run for one epoch using the training documents. A saved adapter is present at results/qwen_cpt_model/checkpoint-epoch-1/adapter_model.safetensors.

Supervised Fine-Tuning (SFT): LoRA-based SFT was run for three epochs using the 304 valid Q/A records (273 training and 31 validation). The recorded final evaluation loss is 2.255685806274414. The saved adapter is results/qwen_sft_model/adapter_model.safetensors.

Retrieval-Augmented Generation (RAG): The project includes src/rag_assistant.py and uses paraphrase-multilingual-MiniLM-L12-v2 embeddings, 700-character chunks, 120-character overlap, top-k=3 and a similarity threshold of 0.55. The initial selected functional test passed 6/6 cases; this is a smoke test, not a comprehensive benchmark.

PROPOSAL 2 — Evaluation and Findings

5. Evaluation Method

The evaluation rubric assigns 1.0 to a fully correct answer, 0.5 to a partially correct answer and 0.0 to an incorrect answer. Weighted score is the sum of question scores, and weighted score percentage is the weighted score divided by 30 and multiplied by 100.

A unified inference run generated 90 predictions: 30 questions each for Base, CPT and SFT, using a shared prompt template, max_new_tokens=150 and do_sample=False. These predictions are stored in results/model_results/phase2_three_model_predictions.csv. The unified review file was initially blank; the CPT-specific manual review was subsequently filled using AI-assisted provisional assessment.

6. Results

Metric	Qwen Base	Qwen CPT	Qwen SFT
Questions	30	30	30
Weighted score	10.5 / 30	9 / 30	13.5 / 30
Weighted score (%)	35%	30%	45%
Fully correct	6	4	7
Partially correct	9	10	13
Incorrect	15	16	10

Important qualification: Base and SFT figures are recovered historical reviewed results. The CPT result (9/30, 30%) is a provisional AI-assisted assessment of 30 predictions and has not been verified by independent human raters. It must not be described as a completed independent human evaluation. CPT category counts are 4 fully correct, 10 partially correct and 16 incorrect.

The Base/SFT comparison shows a descriptive increase of 10 percentage points for SFT over Base (45% versus 35%). SFT had one more fully correct answer and five fewer incorrect answers. These observations are limited to this 30-question reviewed set and do not establish statistical significance or generalization.

7. Error Analysis and Limitations

The reviewed answers show issues with unsupported details, hallucinations, incomplete procedural instructions and incorrect handling of refusal cases. Fine-tuning may improve response behavior, but answer completeness and precise grounding in university-specific instructions remain important areas for improvement. RAG is intended to reduce unsupported answers by retrieving relevant source passages at inference time; its effectiveness depends on retrieval relevance and source coverage.

Limitations: no second independent human rater is recorded; CPT scores are provisional and AI-assisted; the unified 90-prediction review remains distinct from the historical Base/SFT reviewed files; average inference latency and held-out perplexity are not established in the recovered artifacts; and the 6/6 RAG result is only an initial smoke test.

PROPOSAL 2 — Decisions and Handover

8. Decision Log

Decision	Rationale / outcome
Select Qwen2.5-0.5B-Instruct	Chosen as the foundation model; instructor approval is recorded in project history.
Use LoRA for CPT and SFT	Parameter-efficient fine-tuning was used for the available compute environment.
Use document-level 90/10 split	38 source documents split into 34 training and 4 held-out documents, random_state=42.
Prepare separate SFT and test sets	304 valid SFT Q/A records and a separate 30-question evaluation set.
Run three-model inference	30 predictions per model generated with shared prompt and deterministic decoding settings.
Add RAG capability	Multilingual embeddings and retrieval added to ground answers in university support materials.
Report supported metrics only	CPT AI-assisted score is explicitly provisional; unavailable metrics are not estimated.

9. Submission Handover

• GitHub repository: https://github.com/hibanaimat05/JUST-Student-Assistant

• Evaluation artifacts: three-model predictions, Base/CPT/SFT review files and evaluation workbook are committed to the repository.

• Git tag: phase-2 was created and pushed to the remote repository.

• SCOREBOARD.md: updated with the provisional CPT evaluation result and its limitations.

• Demo video: not included because it is no longer required for submission.

• Security: do not commit access tokens or API keys. Revoke temporary GitHub tokens when no longer needed.

Final status: The project includes Qwen CPT and SFT training artifacts, a RAG implementation, recovered Base/SFT reviewed results, a provisional CPT assessment, and a unified 90-prediction inference run. The report distinguishes completed work from provisional evaluation and unavailable metrics.

End of Proposal 2